CHAPTER - 4

ANALYZING FUNCTIONAL DEPENDENCIES,REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The Vilvah Skincare Database System is designed to efficiently manage customers, products, categories, suppliers, inventory, orders, payments and administrators. Functional dependencies identify relationships between attributes, while redundancy analysis and data anomaly detection improve consistency and reduce duplication.

1.CUSTOMER

Functional Dependency

Customer_ID → First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date

Redundancy Analysis

Customer information is stored only once. Orders store only Customer_ID.

Data Anomalies

• Insert Anomaly: A new customer can be added without creating an order.
• Update Anomaly: Updating customer details requires changes only in the Customer table.
• Delete Anomaly: Deleting an order does not remove customer information.

2.CATEGORY

Functional Dependency

Category_ID → Category_Name, Category_Description, Status

Redundancy Analysis

Categories are stored only once.

Data Anomalies

• Insert Anomaly: New categories can be added without products.
• Update Anomaly: Category name is updated only once.
• Delete Anomaly: Deleting a product does not remove the category.

3.PRODUCT

Functional Dependency

Product_ID → Product_Name, Category_ID, Supplier_ID, Price, Weight, Ingredients, Skin_Type, Brand, Expiry_Date

Redundancy Analysis

Product information is stored only once.

Data Anomalies

• Insert Anomaly: Products can be added before orders.
• Update Anomaly: Product details are updated only once.
• Delete Anomaly: Deleting an order does not remove products.

4.SUPPLIER

Functional Dependency

Supplier_ID → Supplier_Name, Contact_Person, Phone, Email, City, GST_Number

Redundancy Analysis

Supplier information is stored only once.

Data Anomalies

- Insert Anomaly: Suppliers can be added independently.
- Update Anomaly: Supplier details are updated once.
- Delete Anomaly: Deleting a product does not remove supplier.

5.INVENTORY

Functional Dependency

Inventory_ID → Product_ID, Quantity, Minimum_Stock, Warehouse, Last_Updated

Redundancy Analysis

Inventory references Product_ID.

Data Anomalies

- Insert Anomaly: Inventory can be maintained independently.
- Update Anomaly: Stock updates affect only Inventory.
- Delete Anomaly: Deleting inventory does not remove product.

6.ORDERS

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date

Redundancy Analysis

Stores only order information.

Data Anomalies

- Insert Anomaly: Orders can be created using Customer_ID.
- Update Anomaly: Order status is updated once.
- Delete Anomaly: Deleting orders does not remove customer.

7.ORDER DETAILS

Functional Dependency

Order_Detail_ID → Order_ID, Product_ID, Quantity, Unit_Price, Discount, Total_Price

Redundancy Analysis

Connects Orders and Products.

Data Anomalies

- Insert Anomaly: Items can be added independently.
- Update Anomaly: Quantity updates affect one record.
- Delete Anomaly: Deleting order details does not remove products.

8.PAYMENT

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Transaction_ID, Payment_Status, Payment_Date, Amount

Redundancy Analysis

Payment is stored separately.

Data Anomalies

• Insert Anomaly: Payments can be added after orders.
• Update Anomaly: Payment status is updated once.
• Delete Anomaly: Deleting payment does not remove order.

9.ADMIN

Functional Dependency

Admin_ID → Admin_Name, Email, Phone, Role, Username, Password

Redundancy Analysis

Admin details are stored separately.

Data Anomalies

• Insert Anomaly: Admins can be added independently.
• Update Anomaly: Admin details are updated once.
• Delete Anomaly: Deleting an admin does not affect business records.

CONCLUSION

The Vilvah database uses proper functional dependencies and normalized tables. Each entity stores only relevant information, minimizing redundancy and preventing insertion, update, and deletion anomalies while maintaining data integrity.